COMP2300 Set 3 Practice Exam

Topics: Sequential Logic, Latches/Flip-Flops, Registers/Memory, FSM, Timing

Time: 120 minutes **Total:** 100 marks

Instructions

- Answer in English.
- Part A is multiple-choice (concept-focused).
- Part B is computational/design-focused. Show key steps.
- Assume positive-edge triggered D flip-flops unless stated otherwise.

Part A: Multiple Choice (30 marks, 3 marks each)

Choose exactly one option for each question.

A1. A sequential circuit differs from a combinational circuit because:

(A) It has no state.
(B) Its output depends only on current inputs.
(C) Its output depends on current inputs and past history.
(D) It cannot be synchronized by a clock.

A2. For a NOR-based SR latch, which input combination is invalid?

(A) S=0, R=0
(B) S=0, R=1
(C) S=1, R=0
(D) S=1, R=1

A3. A D latch is:

(A) Edge-triggered
(B) Level-sensitive
(C) Asynchronous-only
(D) Independent of the clock

A4. A D flip-flop copies D to Q:

(A) Continuously when CLK=1
(B) Only on a clock edge

(C) Only when CLK=0

(D) Only if both S and R are 0

A5. In synchronous sequential circuits, which statement is true?

(A) Cyclic paths are allowed without registers.

(B) State updates are synchronized to clock edges.

(C) Registers may use different clocks freely.

(D) Output logic must be asynchronous.

A6. Addressability refers to:

(A) Number of address bits

(B) Number of memory locations

(C) Bits stored per address

(D) Total capacity in bytes

A7. A Moore FSM output depends on:

(A) Inputs only

(B) State only

(C) State and inputs

(D) Clock frequency only

A8. The aperture time of a flip-flop is:

(A) t_{pcq}

(B) t_{setup}

(C) t_{hold}

(D) $t_{setup} + t_{hold}$

A9. Clock gating can be risky because:

(A) It always improves timing.

(B) It can introduce glitches and delay on the clock.

(C) It eliminates the need for flip-flops.

(D) It converts a D flip-flop into a latch.

A10. One-hot state encoding typically:

(A) Minimizes number of flip-flops.

(B) Maximizes number of flip-flops.

(C) Eliminates output logic.

(D) Requires no next-state logic.

Part B: Computational and Design Problems (50 marks)

B1. SR Latch behavior (8 marks)

An NOR-based SR latch starts with $Q = 0$. For each input pair (S, R) below, determine Q after the latch settles:

$$(0,0),\ (1,0),\ (0,0),\ (0,1),\ (0,0),\ (1,1),\ (0,0)$$

Indicate any invalid/indeterminate cases.

B2. D Latch timing (7 marks)

A D latch starts with $Q = 0$. The table shows CLK and D values over five intervals. Determine Q at the end of each interval.

Interval	1	2	3	4	5
CLK	0	1	1	0	1
D	1	0	1	1	0

B3. D Flip-Flop sampling (6 marks)

A positive-edge D flip-flop samples D at rising edges t_1–t_4 with:

$$D(t_1) = 0,\ D(t_2) = 1,\ D(t_3) = 1,\ D(t_4) = 0$$

If Q starts at 1, list Q after each rising edge.

B4. Registers and memory capacity (9 marks)

A memory has 64K locations and is 16-bit addressable.

(a) How many address bits are required?

(b) What is the total capacity in bits and in bytes?

(c) If the system is byte-addressable instead (8-bit addressability) with the same number of locations, what is the new capacity in bytes?

B5. Timing constraints (10 marks)

A synchronous circuit has $t_{pcq} = 0.4$ ns, $t_{setup} = 0.2$ ns, and the maximum combinational delay $t_{pd} = 6.0$ ns.

(a) Compute the minimum clock period T_c.

(b) Compute the maximum clock frequency f_{max}.

(c) If $t_{hold} = 0.1$ ns, what is the aperture time?

B6. Pipeline throughput (10 marks)

A 3-stage pipeline has stage delays 4 ns, 3 ns, and 5 ns. Flip-flops have $t_{pcq} = 0.3$ ns and $t_{setup} = 0.2$ ns.

(a) Compute the clock period.

(b) How long to process 20 inputs?

Part C: Past Exam Questions (Source-Tagged) (20 marks)

C1. Moore FSM concept (Source: exam24px2.pdf, Q3)
What is a Moore FSM? Discuss with one suitable example.

C2. FSM design (Source: exam24r.pdf, Q4, adapted)
Design a Moore FSM with one input A and one output X. The output X should be 1 if A has been 1 for three consecutive clock cycles; otherwise $X = 0$.

(a) Draw the Moore state transition diagram.

(b) Provide an encoded state transition table.

(c) Derive simplified next-state and output equations.

(d) Draw the high-level FSM schematic.

C3. Pipeline timing (Source: 2023_Final_Exam.pdf, adapted)
A 3-stage pipeline has delays: Stage 1 = 3.4 ns, Stage 2 = 3.3 ns, Stage 3 = 3.7 ns. Flip-flops have $t_{pcq} = 0.2$ ns and $t_{setup} = 0.1$ ns, with $t_{hold} = 0$.

(a) What is the clock period?

(b) How long to process 12 tokens on one pipeline?

(c) How long to process 10^9 tokens on one pipeline?

(d) If four identical pipelines split the work evenly, how long for 10^9 tokens total?